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Kahlert School of Computing
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RESEARCH INTERESTS

General interests: Language design issues regarding proofs, performance, and people. What guarantees do languages offer, how efficiently do they run, and to what extent do they help users meet their goals?

Keywords: Gradual typing, Language design, Formal methods, Human factors

EMPLOYMENT AND EDUCATION

- Assistant Professor of Computer Science, University of Utah July 2023 – ongoing
- Postdoctoral Researcher, Brown University 2021 – 2023
supported by the *CIFellows 2020* program
Mentor: *Shriram Krishnamurthi*
- Ph.D. in Computer Science, Northeastern University 2014 – 2020
Advisor: *Matthias Felleisen*
Thesis: *Deep and Shallow Types*
- M. Eng. in Computer Science, Cornell University 2013 – 2014
Advisor: *Ross Tate*
- Programmer, Rentenna Inc. 2012 – 2014
- B.S. in Industrial and Labor Relations (ILR), Cornell University 2010 – 2013
Minor in Computer Science
- General studies, Hudson Valley Community College 2009 – 2010
toward a guaranteed transfer to Cornell ILR

HONORS AND AWARDS

- Open Source Research Experience: Type Narrowing: A Language Design Benchmark 2025
received summer support for Siva Sathyaseelan, an undergraduate researcher from IIT (BHU) Varanasi
sponsored by the *NSF 2025 Summer of Reproducibility*

- **Open Source Research Experience: Static Python Perf** 2024
received summer support for Mrigank Pawagi, an undergraduate researcher from IIS Bengaluru sponsored by the NSF 2024 Summer of Reproducibility
- **CRA/CCC/NSF CI Fellowship** 2021 – 2023
- SIGPLAN Student Scholarship to **50 Years of the ACM A.M. Turing Award** 2017
- Northeastern CCIS Graduate Community Service Award 2016
- Cornell CS Teaching Award 2014, 2013
- Distinguished Paper Award CAV 2025, ECOOP 2025, Programming 2023
- Distinguished Artifact Award ECOOP 2025

FUNDING

- Price College: AI, Innovation, and Student Success Integration into Required UG Courses
PI Wiese, Co-PIs Greenman, Kopta
- NSF CS2: On-Demand Semantic Analysis via Program Synthesis April 2026
PI Greenman, Co-PIs Regehr, Shankar, D’Antoni (UCSD) **IIS-2546821**
- Price College VPR Seed Grant Competition 2025

PUBLICATIONS

JOURNAL

- Nathaniel Hejduk, Ben Greenman, Matthias Felleisen, and Christos Dimoulas **TOPLAS 2025**
Navigating Mixed-Typed Migration with Profilers
- Ben Greenman, Christos Dimoulas, and Matthias Felleisen. **TOPLAS 2023**
Typed–Untyped Interactions: A Comparative Analysis
- Ben Greenman, Asumu Takikawa, Max S. New, Daniel Feltey, Robert Bruce Findler, Jan Vitek, and Matthias Felleisen. **JFP 2019**
How to Evaluate the Performance of Gradual Type Systems

CONFERENCE & SYMPOSIUM

- Ali Ghanbari, Ben Greenman, Sasan Tavakkol, Shibbir Ahmed **ISSTA 2026**
Provably Lossless Acceleration of DNN Mutation Testing via Memoization

- Xuanyu Peng, Dominic Kennedy, Yuyou Fan, Ben Greenman, John Regehr, Loris D’Antoni
Nice to Meet You: Synthesizing Practical Abstract Transformers POPL 2026
- Ashton Wiersdorf and Ben Greenman
Chorex: Restartable, Language-Integrated Choreographies Programming 10.3, 2026
- Hanwen Guo and Ben Greenman
If-T: A Benchmark for Type Narrowing
Editors’ Choice Award Programming 10.2, 2026
- Siddhartha Prasad, Ben Greenman, Tim Nelson, and Shriram Krishnamurthi
A Misconception-Driven Adaptive Tutor for Linear Temporal Logic
Distinguished Paper Award CAV 2025
- Siddhartha Prasad, Ben Greenman, Tim Nelson, and Shriram Krishnamurthi
Lightweight Diagramming for Lightweight Formal Methods: A Grounded Language Design
Distinguished Paper Award ECOOP 2025
- Ashton Wiersdorf, Stephen Chang, Matthias Felleisen, and Ben Greenman
Type Tailoring ECOOP 2024
- Ben Greenman, Siddhartha Prasad, Antonio Di Stasio, Shufang Zhu, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, and Milda Zizyte
Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic FM 2024
- Tim Nelson, Ben Greenman, Siddhartha Prasad, Tristan Dyer, Ethan Bove, Qianfan Chen, Charles Cutting, Thomas Del Vecchio, Sidney LeVine, Julianne Rudner, Ben Ryjikov, Alexander Varga, Andrew Wagner, Luke West, and Shriram Krishnamurthi
Forge: A Tool and Language for Teaching Formal Methods OOPSLA 2024
- Ben Greenman, Alan Jeffrey, Shriram Krishnamurthi, and Mitesh Shah
Privacy-Respecting Type Error Telemetry at Scale Programming 8.3, 2024
- Siddhartha Prasad, Ben Greenman, Tim Nelson, and Shriram Krishnamurthi
Conceptual Mutation Testing for Student Programming Misconceptions Programming 8.2, 2024
- Siddhartha Prasad, Ben Greenman, Tim Nelson, and Shriram Krishnamurthi
Generating Programs Trivially: Student Use of Large Language Models CompEd, December 2023
- Ben Greenman, Matthias Felleisen, and Christos Dimoulas
How Profilers Can Help Navigate Type Migration OOPSLA 2023
- Matthew Flatt, Taylor Allred, Nia Angle, Stephen De Gabrielle, Robert Findler, Jack Firth, Kiran Gopinathan, Ben Greenman, Siddhartha Kasivajhula, Alex Knauth, Jay McCarthy, Sam Phillips, Sorawee Porncharoenwase, Jens Axel Sogaard, and Sam Tobin-Hochstadt
Rhombus: A New Spin on Macros Without All The Parentheses OOPSLA 2023
- Lukas Lazarek, Ben Greenman, Matthias Felleisen, and Christos Dimoulas
How to Evaluate Blame for Gradual Types, Part 2 ICFP 2023

- Ben Greenman ACM REP, June 2023
GTP Benchmarks for Gradual Typing Performance
- Ben Greenman, Sam Saarinen, Tim Nelson, Programming 7.2, 2023
and Shriram Krishnamurthi
Little Tricky Logic: Misconceptions in the Understanding of LTL
- Kuang-Chen Lu, Ben Greenman, Carl Meyer, Dino Viehland, Programming 7.1, 2023
Aniket Panse, and Shriram Krishnamurthi
Gradual Soundness: Lessons from Static Python
- Siddhartha Prasad, Ben Greenman, Tim Nelson, John Wrenn, Koli Calling 2022
and Shriram Krishnamurthi
Making Hay from Wheats: A Classsourcing Method to Identify Misconceptions
- Ben Greenman PLDI 2022
Deep and Shallow Types for Gradual Languages
- Ben Greenman, Lukas Lazarek, Christos Dimoulas, and Matthias Felleisen Programming 6.2, 2022
A Transient Semantics for Typed Racket
- Kuang-Chen Lu, Ben Greenman, and Shriram Krishnamurthi Programming 6.2, 2022
Types for Tables: A Language Design Benchmark
Editors' Choice Award
- Lukas Lazarek, Ben Greenman, Matthias Felleisen, and Christos Dimoulas ICFP 2021
How to Evaluate Blame for Gradual Types
- Ben Greenman, Matthias Felleisen, and Christos Dimoulas OOPSLA 2019
Complete Monitors for Gradual Types
- Preston Tunnell Wilson, Ben Greenman, Justin Pombrio, Shriram Krishnamurthi. DLS 2018
The Behavior of Gradual Types: A User Study
- Daniel Feltey, Ben Greenman, Christophe Scholliers, Robert Bruce Findler, OOPSLA 2018
and Vincent St. Amour.
Collapsible Contracts: Fixing a Pathology of Gradual Typing
- Ben Greenman, Matthias Felleisen. ICFP 2018
A Spectrum of Type Soundness and Performance
- Ben Greenman, Zeina Migeed. PEPM 2018
On the Cost of Type-Tag Soundness
- Sam Tobin-Hochstadt, Matthias Felleisen, Robert Bruce Findler, Matthew Flatt, SNAPL 2017
Ben Greenman, Andrew M. Kent, Vincent St-Amour, T. Stephen Strickland,
and Asumu Takikawa.
Migratory Typing: 10 Years Later
- Stephen Chang, Ben Greenman, and Alex Knauth. POPL 2017
Type Systems as Macros

- Asumu Takikawa, Daniel Feltey, Ben Greenman, Max S. New, Jan Vitek, and Matthias Felleisen. *Is Sound Gradual Typing Dead?* POPL 2016
- Ben Greenman, Fabian Muehlboeck, and Ross Tate. *Getting F-Bounded Polymorphism into Shape* PLDI 2014

WORKSHOP

- Shriram Krishnamurthi, Siddhartha Prasad, Tim Nelson, and Ben Greenman *A Tutor for Linear Temporal Logic* TEAL@FLOC 2026
- Ashton Wiersdorf and Ben Greenman *Bushwhacking Toward Event-Driven Chorex* Choreo 2026
- Dibri Nsofor and Ben Greenman *Toward a Corpus Study of the Dynamic Gradual Type* HATRA 2024
- Taylor Allred, Xinyi Li, Ashton Wiersdorf, Ben Greenman, and Ganesh Gopalakrishnan *FlowFPX: Nimble Tools for Debugging Floating-Point Exceptions* JuliaCon 2023
- Asumu Takikawa, Daniel Feltey, Ben Greenman, Max S. New, Jan Vitek, and Matthias Felleisen. *Position Paper: Performance Evaluation for Gradual Typing* STOP 2015

INVITED TALKS

- **Formally Speaking Seminar at Code Metal** September 2026
Typing Python, Gradually
- **WITS** January 2026
Type Narrowing the Hard Way
- Amazon Compilers Tech Talk December 2025
Nice to Meet You: Synthesizing Practical MLIR Transformers
- RPI CS Seminar June 2025
Kicking the Ladder Away: From Gradual Types to Plain Types
- **Iowa State CS Colloquium** November 2024
Toward a Science of Type System Design
- **Research Challenges in Computing @ University of Utah** 2024
Rigorous Methods for Language Design
- **PLT @ Northwestern University** September 2024
Teaching Formal Methods with Forge
- **IETF 120: Usable Formal Methods Research Group** July 2024
Forge: Usable Model-Finding

- **BYU Grad Seminar** November 2023
How Profilers Can Help Navigate Type Migration
- **TLf@AAAI-SSS'23** March 2023
Towards LTLf Misconceptions
- **VardiFest** 2022
NJPLS
Little Tricky Logic: Misconceptions in the Understanding of LTL
- **Racket Con** 2020, 2022
Shallow Typed Racket
Shallow and Optional Types for Typed Racket
- **Boston University POPV Seminar** 2020
Complete Monitoring for Gradual Types
- **GRACE Workshop** 2018
Three Approaches to Gradual Typing

TEACHING

UTAH

			Enrollment (Responded)	Course (Avg)	Instructor (Avg)
Spring 25	CS 6110	Software Verification	TBD	TBD (5.09)	TBD (5.20)
Spring 25	CS 7936	Ph.D Seminar	TBD	TBD (5.09)	TBD (5.20)
Fall 25	COMP 1020	Programming for All 2	TBD	TBD (5.17)	TBD (5.31)
Spring 25	CS 4470	Compilers	58 (51)	5.28 (5.24)	5.43 (5.28)
	CS 7936	Ph.D Seminar	6 (?)	6	6
Fall 24	N/A	<i>parental leave</i>			
Spring 24	CS 5110/6110	Software Verification	22 (20)	5.5 / 5.82 (5.18)	6 / 5.68 (5.21)
Fall 23	CS 3520/6520	Programming Languages	159 (77)	5.32 / 5.82 (5.12)	5.45 / 5.68 (5.19)

BROWN

- Topics in PL and Systems: Tables and Humans 2021
Seminar Organizer & Scribe

NORTHEASTERN

- Software Development 2018, 2020
Teaching Assistant
- Fundamentals I 2016
Teaching Assistant
- Object-Oriented Design 2016
Teaching Assistant

CORNELL

- Functional Programming and Data Structures
Teaching Assistant

2012 – 2014

ADVISING

PH.D.

- Ashton Wiersdorf, started Fall 2023
- Dominic Kennedy, started Fall 2024
- Hanwen Guo, started Fall 2024

MASTERS

- Dibri Nsofor, MSc
project: *Data Science for Gradual Types*
- Suyasha Bobhate, IS Fall 2023
project: *Quantum Key-Value Stores*

expected Spring 2026

graduated Spring 2024

UNDERGRAD

- Jackson Brough, BS
thesis: *Constructive Real Analysis via Locators*

expected Spring 2026

COMMITTEE MEMBERSHIP

- Zhaofeng Li, Ph.D, advisor [Anton Burtsev](#)
- Sara Nurollahian, Ph.D, advisor [Eliane Wiese](#)

INFORMAL MENTEES

Siva Sathyaseelan		IIT (BHU) Varanasi	Summer 2025
Mrigank Pawagi		IIS Bengaluru	Summer 2024
Vivaan Rajesh		Hillcrest High School	2023 – 2024
Siddhartha Prasad	Ph.D.	Brown University	2022 – ongoing
Rob Durst			Fall 2023
Caspar Popova			Spring – Fall 2023
Aniket Karna	M.S.	University of Utah	Fall 2023
Taylor Allred	M.S.	University of Utah	2022 – 2023
Qianfan Chen	Sc.B.	Brown University [thesis]	2021 – 2022
Kuang-Chen Lu	Ph.D.	Brown University	2021 – 2022
Milo Davis	B.S.	Northeastern University	2017
Zeina Migeed	B.S.	Northeastern University	2016 – 2017

DEPARTMENT, COLLEGE, AND UNIVERSITY SERVICE

- Committee Member: Lecturing Faculty Hiring Fall 2025 – Spring 2026
- Faculty Mentor: CS 1960: Success in Computing Summer 2025 – ongoing
- Committee Member: Graduate Admissions Spring 2026, 2025
- Teacher: Price College Hi-Gear Summer Camp Summer 2026, 2025
- Teacher: Price College Exploring Engineering Summer Camp Summer 2024
- Teaching Area Chair: Programming Languages and Web Fall 2023 – ongoing
- Committee Member: K-12 Outreach Planning Committee Fall 2023 – Summer 2025

EXTERNAL SERVICE

- Co-Chair of Workshop Organization ICFP 2026, ICFP/SPLASH 2025
- Co-Chair of Artifact Evaluation Committee & ERC OOPSLA 2023, 2022
- Program Committee DLS 2022
HATRA 2025, 2024, 2023, 2022
ICFP 2026, 2021
OOPSLA 2025
POPL 2027
PLDI 2025, 2021
Scheme 2025
SOAP 2024
TFP 2026, 2025, 2023
- External Review Committee ESOP 2023, ICFP 2023, OOPSLA 2026
- Journal Review JFP 2026, 2024, 2023, 2020, 2019
JuliaCon 2024
SoftwareX 2025
STTT 2024
TOPLAS 2025, 2023
- NSF Panel Review 2025, 2024
- Artifact Evaluation Committee ECOOP 2017; OOPSLA 2017, 2016
- Session Chair ICFP 2026, 2021; NJPLS 2023; OOPSLA 2023; POPL 2026; Programming 2026
- SIGPLAN-M Long-Term Mentor Fall 2024 – ongoing
- El Turco: Human-AI dialogue Spring 2024
show: Mori Art Museum, 2025-02-13 – 2025-06-08
- Senior Division Judge: University of Utah Science and Engineering Fair (USEF) 2026, 2025

PROFESSIONAL MEMBERSHIPS

- IEEE, Member 2023 – ongoing
- IEEE Computer Society, Member 2023 – ongoing
- ACM, Member 2023 – ongoing
- ACM SIGPLAN, Member 2016 – ongoing
- Sigma Xi, Member 2025
The Scientific Research Honor Society
- Phi Theta Kappa, Member 2013
2-year college Honor Society

BIOGRAPHY

Ben Greenman is an assistant professor in the Kahlert School of Computing at the University of Utah. He earned his PhD from Northeastern University in 2020 under the supervision of Matthias Felleisen. He was a CIFellows 2020 postdoc at Brown University mentored by Shriram Krishnamurthi. His research seeks to elevate language design from an art to a science by systematic methods that assess formal guarantees, performance, and human factors concerns.